

Model Question Paper I
ENERGY CONSERVATION AND AUDIT

*Time: 3 Hour**Max. Marks: 75***PART A**

- I. Answer **all** questions in one word or one sentence. Each question carries 1 mark.
 (9 * 1 = 9 marks)

1	Name the energy that is extracted or captured directly from natural resources.	M1.01	R
2	Define energy intensity ratio.	M1.01	R
3	Name the type of Lamp which is an energy efficient replacement for Fluorescent Lamp	M2.01	R
4	What type of cores are used in electrical distribution transformers in providing excellent opportunity to conserve energy right from the installation?	M2.03	R
5	Define cogeneration.	M3.01	R
6	What is the main process occurred in the topping cycle of cogeneration?	M3.02	R
7	What is the key to a systematic approach for decision-making in the area of energy management?	M4.01	R
8	Which instrument has in-built chemical cells which measure various gases such as CO ₂ , CO, NO _x , SO _x etc?	M4.02	R
9	What is a preliminary energy audit?	M4.01	R

PART B

- II. Answer any **eight** questions from the following, each question carries 3 marks.
(8*3=24marks)

1	Define energy security. State the basic aim for implementing energy security for our nation?	M1.01	R
2	List any four strategies that can be used to meet future energy requirements.	M1.02	R
3	Explain the Role of State Designated Agencies as per Energy Conservation Act 2001.	M1.03	U
4	List the advantages of energy efficient motors.	M2.02	R
5	Outline any six General Energy Saving Opportunities with lighting system.	M2.01	U
6	Explain about energy efficient transformers.	M2.03	U
7	Define the term co-generation and state its necessity.	M3.03	R
8	List any three Prime Movers for Cogeneration and give a brief outline about each.	M3.04	R
9	Define energy audit as per EC act 2001 and state its need.	M4.01	R
10	Explain what is meant by targeted energy audit.	M4.01	U

PART C

Answer ALL questions. Each question carries 7 marks.

(6*7=42 marks)

III	List any seven benefits of applying energy conservation in industry.	M1.02	R												
	OR														
IV	Solve and estimate the payback period if the electricity charges are rupees 7 per kwh for a non-Star Rated Refrigerator to be replaced with Star rated Refrigerator (220 Liters). Assume following details	M1.03	A												
	<table><tr><td></td><td>Star rated device</td><td>Non star rated device</td></tr><tr><td>Cost</td><td>Rs 34,000/-</td><td>Rs 20,000/-</td></tr><tr><td>Star level</td><td>5 star</td><td>No star</td></tr><tr><td>Annual Electricity consumption</td><td>816 Units</td><td>2080Units</td></tr></table>		Star rated device	Non star rated device	Cost	Rs 34,000/-	Rs 20,000/-	Star level	5 star	No star	Annual Electricity consumption	816 Units	2080Units		
	Star rated device	Non star rated device													
Cost	Rs 34,000/-	Rs 20,000/-													
Star level	5 star	No star													
Annual Electricity consumption	816 Units	2080Units													
V	Outline the salient features of Indian Electricity Act 2010.	M1.03	U												
	OR														
VI	Explain Bachat Lamp Yojana.	M1.04	U												

VII	A 3-phase AC induction motor (20 kW capacity) is used for pumping operation. Electrical parameters such as current, volt and power factor were measured with power analyzer. (Volts. = 440 V, current = 25 amps and PF = 0.90). Calculate the energy saving in kwh by replacing this ordinary motor with efficiency 80 % by an energy efficient motor with efficiency 90% for one year. Assume that the motor operates 5000hrs per annum. OR	M2.02	U
VIII	List any six energy management techniques in electric motors	M2.02	U
IX	Draw the schematic diagram of back pressure steam turbine cogeneration system and explain it OR	M3.02	U
X	Explain the factors governing the selection of a cogeneration system.	M3.03	U
X1	Explain the principle and need of cogeneration OR	M3.01	U
X11	What are the main classifications of the cogeneration system?	M3.02	R
XIII	Identify the Ten Steps Methodology for Conducting Detailed Energy Audit (as per BIS standard) OR	M4.02	A
XIV	List any seven Instruments used for energy audit and mention their purpose.	M4.04	U

BLUE PRINT

Mark Distribution

Mod ule	Hr / Mod ule	(hi / ΣHi) * 123	TYPE OF QUESTIONS							
			PART A		PART B		PART C		TOTAL	
			No of Question s	Marks	No of Question s	Marks	No of Question s	Marks	No of Question s	Marks
I	19		2		3		4		9	
		32.0 1		2		9		28		39
II	18		2		3		2		7	
		30.3 3		2		9		14		25
III	18		2		2		4		8	
		30.3 3		2		6		28		36
IV	18		3		2		2		7	
		30.3 3		3		6		14		23
Total	73		9		10		12		31	
		123		9		30		84		123

Cognitive Level Wise Question Analysis

Mark Distribution

Cogn itive Level	% Mar ks	Mar ks	TYPE OF QUESTIONS							
			PART A		PART B		PART C		TOTAL	
			No of Question s	Marks	No of Question s	Marks	No of Question s	Marks	No of Question s	Marks
R	30		9		6		2		17	
		36.9		9		18		14		41
U	50		0		4		8		12	
		61.5		0		12		56		68
A	20		0		0		2		2	
		24.6		0		0		14		14
Total	100		9		10		12		31	
		123		9		30		84		123

Question Wise Analysis

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.01	R	1	2
I.2	M1.01	R	1	2
I.3	M2.01	R	1	2
I.4	M2.03	R	1	2
I.5	M3.01	R	1	2
I.6	M3.02	R	1	2
I.7	M4.01	R	1	2
I.8	M4.02	R	1	2
I.9	M4.01	R	1	2
II.1	M1.01	R	3	7
II.2	M1.02	R	3	7
II.3	M1.03	U	3	8
II.4	M2.02	R	3	8
II.5	M2.01	U	3	7
II.6	M2.03	U	3	8
II.7	M3.03	R	3	7
II.8	M3.04	R	3	7
II.9	M4.01	R	3	7
II.10	M4.01	U	3	7
III.	M1.02	R	7	17
IV.	M1.03	A	7	17
V	M1.03	U	7	17
VI	M1.04	U	7	17
VII	M2.02	U	7	17
VIII	M2.02	U	7	17

IX	M3.02	U	7	17
X	M3.03	U	7	17
XI	M3.01	U	7	17
XII	M3.02	R	7	17
XIII	M4.02	A	7	17
XIV	M4.03	U	7	17
Total			123	295

Model Question Paper II
ENERGY CONSERVATION AND AUDIT

Time: 3 Hour

Max. Marks: 75

PART A

I. Answer **all** questions in one word or one sentence. Each question carries 1 mark.
(9 * 1= 9 marks)

1	List the two responsibilities SDA.	M1.03	R
2	Define MEPS.	MI.04	R
3	State the importance of star labelling in devices.	M1.04	R
4	Which is the best suited location to install APFC panel?	M2.04	R
5	Summarize the two needs of cogeneration.	M3.01	U
6	List two prime movers for cogeneration.	M3.04	R
7	Define energy audit as per EC act 2001.	M4.01	R
8	Explain the purpose of manometer.	M4.02	U
9	How Sankey diagram helps in energy analysis?	M4.04	R

PART B

III. Answer any **eight** questions from the following, each question carries 3 marks.
(8*3=24 marks)

1	Identify various stages of an energy pyramid.	M1.02	A
2	Summarize Demand side management.	M1.03	U
3	Outline the need of soft starters in Induction motor starting.	M2.04	U
4	List any six energy management techniques in electric motors.	M2.02	R

5	Draw and explain IPFC.	M2.04	U
6	Draw the Schematic Diagram of Gas Turbine Cogeneration system and explain	M3.02	U
7	Explain bottoming cycle with diagram.	M3.03	U
8	List the needs of energy audit.	M4.02	R
9	Explain the types of energy audit.	M4.02	U
10	Illustrate the Sankey diagram and explain.	M4.04	U

PART C

Answer ALL questions. Each question carries 7 marks.

(6*7=42 marks)

III	Outline the salient features of Indian Electricity Act 2010.	M1.03	U
	OR		
IV	Explain energy conservation and the need of energy conservation.	M1.02	U
V	Explain the Role of State Designated Agencies as per Energy Conservation Act 2001.	M1.03	U
	OR		
VI	Explain Bachat Lamp Yojana.	M1.04	U
VII	Identify various energy saving practices in Lighting	M2.01	A
	OR		
VIII	Draw and explain two energy conservation equipment.	M2.04	U
IX	Summarize energy efficient motors and the techniques for improving the efficiency of energy efficient motors.	M2.02	U
	OR		
X	Explain energy efficient transformers.	M2.03	U

XI	Draw and explain two types of topping cycle cogeneration systems.	M3.02	U
	OR		
XII	Identify the factors governing cogeneration system.	M3.03	A
XIII	Explain about energy audit for building.	M4.03	U
	OR		
XIV	What is the typical energy audit reporting format?	M4.02	R

BLUE PRINT

Mark Distribution

Mod ule	Hr / Mod ule	(hi / Σ Hi) * 123	TYPE OF QUESTIONS							
			PART A		PART B		PART C		TOTAL	
			No of Question s	Marks	No of Question s	Marks	No of Question s	Marks	No of Question s	Marks
I	19	32.0 1	2		2		4		8	
				2		6		28		36
II	18	30.3 3	2		3		4		9	
				2		9		28		39
III	18	30.3 3	2		2		2		6	
				2		6		14		22
IV	18	30.3 3	3		3		2		8	
				3		9		14		26
Total	73	123	9		10		12		31	
				9		30		84		123

Cognitive Level Wise Question Analysis

Mark Distribution

Cognitive Level	% Marks	Marks	TYPE OF QUESTIONS							
			PART A		PART B		PART C		TOTAL	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
R	30		7		2		1		10	
		36.9		7		6		7		20
U	50		2		7		9		18	
		61.5		2		21		63		86
A	20		0		1		2		3	
		24.6		0		3		14		17
Total	100		9		10		12		31	
		123		9		30		84		123

Question Wise Analysis

Q. No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.03	R	1	2
I.2	M1.04	R	1	2
I.3	M1.04	R	1	2
I.4	M2.04	R	1	2
I.5	M3.01	U	1	2
I.6	M3.04	R	1	2
I.7	M4.01	R	1	2
I.8	M4.02	U	1	2
I.9	M4.04	R	1	2
II.1	M1.02	A	3	7
II.2	M1.03	U	3	7
II.3	M2.04	U	3	8
II.4	M2.02	R	3	7

II.5	M2.04	U	3	8
II.6	M3.02	U	3	7
II.7	M3.03	U	3	8
II.8	M4.02	R	3	7
II.9	M4.02	U	3	7
II.10	M4.04	U	3	7
III.	M1.03	U	7	17
IV.	M1.02	U	7	17
V	M1.03	U	7	17
VI	M1.04	U	7	17
VII	M2.01	A	7	17
VIII	M2.04	U	7	17
IX	M2.02	U	7	17
X	M2.03	U	7	17
XI	M3.02	U	7	17
XII	M3.03	A	7	17
XIII	M4.03	U	7	17
XIV	M4.02	R	7	17
Total			122	293

Scoring Indicators**Model Question Paper I****ENERGY CONSERVATION AND AUDIT**

Q No	Scoring Indicators	Split score	Sub Total	Total Score
	PART A			
I. 1	Renewable	1	1	1
I. 2	Gross Inland energy consumption (GIEC) and Gross Domestic Product (GDP)	1	1	1
I. 3	LED	1	1	1
I. 4	Amorphous metal	1	1	1
I. 5	It is defined as the sequential generation of two different forms of useful energy from a single primary energy source, typically mechanical energy and thermal energy	1	1	1
I. 6	In topping cycle of cogeneration, the fuel supplied is used to first produce power and then thermal energy.	1	1	1
I. 7	Energy Audit	1	1	1
I. 8	Energy Audit Instruments Combustion Analyzer	1	1	1
I. 9	Preliminary Energy audit takes 1- 3 days and Performed by a quick walk through the facilities and by analyzing the utility and fuel bills.	1	1	1

	PART B			
1	Define energy security. State the basic aim for implementing energy security for our nation ? <i>Definition (1.5)</i> <i>Basic aim (1.5)</i>	1.5+ 1.5	3	3
	Energy security is defined as “The continuous availability of energy in varied forms in sufficient quantities at reasonable prices”. The basic aim of energy security for a nation is to reduce its dependency on the imported energy sources for its economic growth. Any disruption in energy supplies would be harmful to the country’s economic growth, human survival and wellbeing. The energy import dependence implies vulnerability to external price shocks and supply fluctuations, which in turn threaten the energy security of the country.			
2	List any four strategies that can be used to meet future energy requirements <i>Any 3 strategies (3 x 1 mark)</i> Some of the strategies that can be used to meet future energy requirements include: a. Reducing energy requirements b. Developing renewable energy sources especially solar and wind c. Substituting imported oil/gas with domestic alternatives d. Diversifying energy supply sources	3 x1	3	3

	Mix of fuel comprising of coal, gas, nuclear, hydro and renewables with no dependence on any particular fuel e. Expanding energy resource and developing alternative energy sources Improved Oil Recovery (IOR) and Enhanced Oil Recovery (EOR) for improving exploitation of reserves - Recovery of oil and gas from abandoned or marginal fields - In-situ coal gasification - Capturing Coal Bed Methane (CBM) which escapes from coal seams during mining - Promoting Community Biogas Plants - Energy plantations			
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3	State the Role of State Designated Agencies as per Energy Conservation Act 2001 <i>Any 3role (3x 1 mark)</i>	3 x 1	3	3
	Role of State Designated Agencies As per Energy Conservation Act 2001, State Governments have been empowered to designate agencies (State Designated Agency, i. e. SDA) in consultation with the Bureau of Energy Efficiency. Designated Agencies will have the responsibility to implement the Act within the State. <i>Responsibilities</i> ● Spread awareness on EC Act ● Undertake voluntary initiatives to promote energy conservation ● Liaison and coordinate with BEE, State Government Departments dealing with energy, industry, planning, regulators, consumer affairs, municipal bodies etc. ● Capacity building of staff employed ● Launch and maintain state specific website addressing the voluntary and mandatory provisions of EC Act ● Undertake energy conservation awareness program for consumers, industrial & commercial sector, school children, farmers etc. ● Arrange interactive meets between energy managers, energy auditors, designated consumers and other experts			

4	List the advantages of an energy efficient motor. <i>List 6 (6 X .5Mark)</i>	6x.5	3	3
	● The EEM has greater efficiency than a standard motor. ● EEM has lower slip so they have a higher speed than standard motor ● It can reduce maintenance cost and improve operations in industry due to robustness and reliability ● Operate more satisfactory under abnormal voltage ● Electric power saving ● Operating temperature less ● Noise level is lower			
5	Outline any six General Energy Saving Opportunities with lighting system <i>Any 6 suggestion with one line explanation (6 x .5 mark)</i>	6 x .5	3	3
	● Clean the luminaries by removing dust and dirt ● Daylight Utilization ● Improve room Colour and reflexivity ● Improved Lighting Controls by timers & sensors ● Use More efficient lamps ● Use More efficient accessories-lenses, shades, reflectors ● Task lighting			

6	Explain about energy efficient transformers. <i>Any 3points(3x1)</i>			
	• Energy efficient transformers use an amorphous core- with unique physical and magnetic properties. • These transformers have increased efficiency even at low loads - 98.5% efficiency at 35% load. • It provides an excellent opportunity to conserve energy right from the installation. • These transformers are a little costlier than conventional iron core transformers, • The expected reduction in core loss over conventional (Si Fe core) transformers is roughly around 70%, which is quite significant.	3x1	3	3
7	Define the term cogeneration and its necessity <i>Definition: 1 mark+ any two needs: 2 mark</i>	1+2	3	3

	● Cogeneration or Combined Heat and Power (CHP) is defined as the sequential generation of two different forms of useful energy from a single primary energy source, typically mechanical energy and thermal energy. ● Mechanical energy may be used either to drive an alternator for producing electricity, or rotating equipment such as motor, compressor, pump or fan for delivering various services. ● Thermal energy can be used either for direct process applications or for indirectly producing steam, hot water, hot air for dryers or chilled water for the process cooling. ● Cogeneration provides a wide range of technologies for application in various domains of economic activities. The overall efficiency of energy use in cogeneration mode can be up to 85 per cent and above in some cases. ● Along with the saving of fossil fuels, cogeneration also allows to reduce the emission of greenhouse gases (particularly CO₂ emission). ● The production of electricity being on-site, the burden on the utility network is reduced and the transmission line losses eliminated.			
8	List any three Prime Movers for Cogeneration and give a brief outline about each. <i>Any 3(3x.5) + Explanation each (3x.5)</i>			

	● Steam turbines. ● Gas turbines. ● Reciprocating engines. ● Fuel cells ● Organic Rankine cycles, Steam Turbine Steam turbines come in a huge range of sizes, with the smallest being 80kW and the largest being 500,000kW. Gas Turbine Gas turbines are another prime mover choice, coming in sizes from 1,000 to 30,000kW. Spark Ignition Reciprocating Engine Spark ignition reciprocating engines come in sizes ranging from 50 to 4,000kW and are a quite common prime mover choice. Fuel Cell A fuel cell is a smaller prime mover, ranging in size from 5 to 1,400kW. Microturbine Microturbines, as their name suggest, come in smaller sizes from 30 to 250kW Organic Rankine Cycle Organic Rankine cycle comes in sizes from 30 to 10,000kW.	3x.5+ 3x.5	3	3
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9	Define energy audit as per EC act 2001 and state its need. <i>Definition: 1 mark+ any two needs: 2 mark</i> “Energy Audit” means the verification, monitoring and analysis of use of energy including submission of technical report containing recommendations for improving energy efficiency with cost benefit analysis and an action plan to reduce energy consumption. • The Energy Audit provides the vital information base for overall energy conservation program covering essentially energy utilization analysis and evaluation of energy conservation measures. • It provides recommendations for improving energy efficiency with cost benefit analysis and an action plan to reduce energy consumption. • Energy audit will help to understand more about the ways energy is used in the industry, and help in identifying the areas where waste can occur and where scope for improvement exists. • In general, energy audit is the translation of conservation ideas into realities, by evolving technically feasible solutions with economic and other organizational considerations within a specified time.	1+2	3	3
10	Explain what is meant by targeted energy audit. <i>Definition and explanation</i> <i>(1.5+1.5)</i>			

	Targeted Energy Auditing: ● Based on the outcome of the preliminary audit result. It Provides data and analysis on specified target projects. ● An organization may target its lighting system or boiler system or compressed air system with a view to bring out energy savings. ● Targeted audits therefore involve detailed surveys of the target subjects/areas with analysis of energy flows and costs associated with those targets.	1.5+ 1.5	3	3
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	PART C			
III	List any seven benefits of applying energy conservation in industry <i>Any seven (7x 1Marks)</i>	7 X 1	7	7
	● Reduced energy bills ● Increased competitiveness ● Increased productivity ● Improved quality ● Increased profits ● Maintains sustainable environment ● Reduced raw materials cost			

IV

Estimate the payback period if the electricity charges are rupees 7 per kwh for a non-Star Rated Refrigerator to be replaced with Star rated Refrigerator (220 Liters). Assume following details

	Star rated device	Non star rated device
Cost	Rs 34,000/-	Rs 20,000/-
Star level	5 star	No star
Annual Electricity consumption	816 Units	2080Units

Annual energy savings (4) +Payback period (3)

4+3

7

7

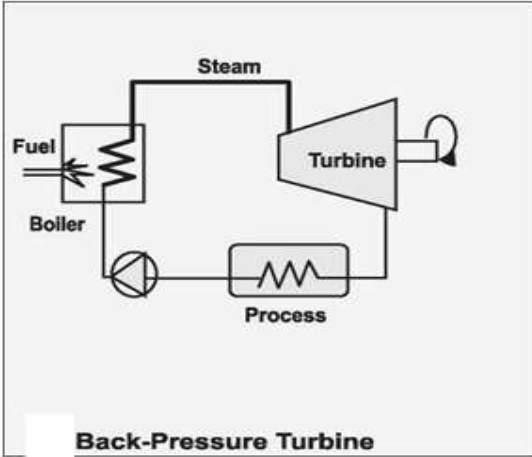
	Star rated device	Non star rated device
Cost	Rs 34,000/-	
Star level	5 star	No star
Annual Electricity Consumption	816 Units	2080Units
Annual energy savings	2080-816 = 1264 Units	
Annual savings in rupees @ rupees 7/=	1264 x 7 = 8848	
Payback period =	34000/8848 = 3. 84 = 4 Years	

V	<i>Any 7 points(7x1) marks</i> The salient features of energy conservation act 2001 are EC act empowers the union government and in some instances the state government to ● Notify energy-intensive industries, establishments and commercial buildings as designated consumers ● Prescribe energy consumption norms and standards for designated consumers ● Direct designated consumers to appoint certified energy managers for efficient use of energy ● State government to amend the energy conservation building codes to suit regional and climatic conditions ● Direct owners of commercial building to comply with the energy conservation building codes ● Direct mandatory display of labels on notified equipment and appliances ● Specify energy consumption standards for notified equipment and appliances ● Prohibit manufacture, sale, purchase and import of notified equipment and appliances not conforming to standards ● Under the provision of this Act the Bureau of Energy Efficiency (BEE) was established with effect from 1.03.2002 ● The mission of the BEE is to institutionalize energy efficiency services, promote energy efficiency delivery mechanisms and provide leadership for improvement of energy efficiency in all sectors of the economy.	7x1	7	7
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VI	<i>Explanation (2) +benefits(5)</i> The total installed generating capacity in the country is over 1,56,700 Approx 28000 MW (i.e. 17 %) is lighting load. There are no mandatory policies or regulations for the adoption of CFLs in households in India. Some of the progressive utilities in India have tried out bulk procurement and marketing of CFLs to households at below market prices and / or introducing replacement schemes. –could not sustain. National Mission for Enhanced Efficiency (NMEEE) under the National Action Plan for Climate Change (NAPCC) also prompts the BEE to accelerate the shift towards energy efficient appliances. The Bachat Lamp Yojana (BLY) scheme is an indirect outcome of the above policy scenario. Hence, the BLY objective is to – replace ICLs amongst residential users in India with quality long-life CFLs. Benefits ● Reduced Peak Load Demand, less load shedding and power-cuts. ● Reduced installation ratings, thus lowering peak demand reduces overall plant and Capital cost requirements. ● Reduced Electricity Bills ● Less Load shedding ● High Returns on the investment from the CERs generated.	2+5	7	7
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VII	A 3-phase AC induction motor (20 kW capacity) is used for pumping operation. Electrical parameters such as current, volt and power factor were measured with power analyser. (Volts. = 440 V, current = 25 amps and PF = 0.90). Calculate the energy saving in kwh by replacing this ordinary motor with efficiency 80 % by an energy efficient motor with efficiency 90% for one year. Assume that the motor operates 5000hrs per annum Ordinary motor energy(3)+EEM(4)			
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	$P_{out} = \sqrt{3} V_L I_L \cos\phi$ $= \sqrt{3} 440 \times 25 \times 0.9$ $= 19030 \text{ W}$ <u>Case 1 : Ordinary motor</u> $P_{in} = P_{out}/\eta$ $= 19030/.8$ $= 23787.5 \text{ W}$ Energy consumed by Ordinary motor in one year $E = P_{in} \times \text{Time}$ $E = 23787 \times 5000 = 118935000 \text{ wh} = 118935 \text{ kwh}$ <u>Case 2 : Energy efficient motor</u> $P_{in} = P_{out}/\eta$ $= 19030/.9$ $= 21144.44 \text{ W}$ Energy consumed by energy efficient motor in one year $E = P_{in} \times \text{Time}$ $E = 21144.44 \times 5000 = 105722200 \text{ wh}$ $= 105972.2 \text{ kwh}$ Energy Saving in one year = $118935 - 105972.2$ $= 12962.8 \text{ kwh}$	3+4	7	7
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VIII	<i>Any 7 points(7x1) marks</i> • Improve Quality of power supply • Improve controls • Improve Power factor • Improve cooling • Turn motors off if not in use • Schedule Regular maintenance • Provide Regular lubrication • Reduce Peak Demand by rescheduling motor operation	7x1	7	7
IX	<i>Schematic diagram (4 mark)</i> <i>Explanation (3 marks)</i>  Back-Pressure Turbine	4 + 3	7	7

	• This system is employed where the end-user needs thermal energy at two different temperature levels. • Back pressure refers to steam turbines that exhaust steam at above atmospheric pressures used for drive application (Sugar mill drives, Sugar fibrizor / Shredder drive, Pump drives). • The power generation is incidental to the process of steam demand. • The back pressure turbine with one controlled extraction point is possible, this extraction steam is also used to meet process steam demand at intermediate pressure when volume demand is high and pressure variations cannot be tolerated. • These turbines are of extraction cum back pressure type. • The back pressure turbine may also have bleed points (uncontrolled extractions) to satisfy steam demands at intermediate pressures. This provision is applicable when the bleed (medium pressure) steam volume demand is low and pressure variations can be tolerated. These turbines are of bleed cum back pressure type.			
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X	<i>Any 7 points(7x1) marks</i> Electrical load matching: - In the base electrical load matching, size of cogeneration plant is chosen to fulfill minimum demand curve. The remaining power need is satisfied by purchasing from the utility. Thermal load matching: - In base thermal load matching, minimum heat requirement of the premises is considered while sizing of the cogeneration plant. auxiliary boilers can supply heat in case of additional heating needs. If excess thermal energy is available, it can be exported to the needy places. Thermal to electrical energy ratio: - This is also called as heat to power ratio. - Heat to power ratio is one of the most influential technical parameters which can govern the selection of cogeneration system. Quality of required thermal energy: - Quality of thermal energy refers to the temperature and pressures of the steam which can be an attributing factor for the selection of cogeneration scheme. Fuel availability: - The type of readily available fuel is also an important consideration while choosing a cogeneration system. Almost every fuel can be used by a system having compatibility with the same. Installation and available space: - Cogeneration system may depend upon whether it's a retrofit or new installation. Grid-connected/independent system: - Technical design of cogeneration plant may depend upon whether its connected to grid or it's a standalone system. Operational costs:	7	7	7
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XI	<i>Definition (2) +needs (5)</i> ● Cogeneration or Combined Heat and Power (CHP) is defined as the sequential generation of two different forms of useful energy from a single primary energy source, typically mechanical energy and thermal energy. ● Mechanical energy may be used either to drive an alternator for producing electricity, or rotating equipment such as motor, compressor, pump or fan for delivering various services. ● Thermal energy can be used either for direct process applications or for indirectly producing steam, hot water, hot air for dryers or chilled water for the process cooling. ● Cogeneration provides a wide range of technologies for application in various domains of economic activities. The overall efficiency of energy use in cogeneration mode can be up to 85 per cent and above in some cases. ● Along with the saving of fossil fuels, cogeneration also allows to reduce the emission of greenhouse gases (particularly CO2 emission). ● The production of electricity being on-site, the burden on the utility network is reduced and the transmission line losses eliminated.	2+5	7	7

XII	<i>Classification (1) +Explanation(bottoming(2) +Topping(4))</i> Cogeneration systems are normally classified according to the sequence of energy use and the operating schemes adopted. • A cogeneration system can be classified as either a topping or a bottoming cycle on the basis of the sequence of energy use. In a topping cycle, the fuel supplied is used to first produce power and then thermal energy, which is the by-product of the cycle and is used to satisfy process heat or other thermal requirements. Topping cycle cogeneration is widely used and is the most popular method of cogeneration. Bottoming Cycle • In a bottoming cycle, the primary fuel produces high temperature thermal energy and the heat rejected from the process is used to generate power through a recovery boiler and a turbine generator. Bottoming cycles are suitable for manufacturing processes that require heat at high temperature in furnaces and kilns, and reject heat at significantly high temperatures. Topping cycle • A gas turbine or diesel engine producing electrical or mechanical power followed by a heat recovery boiler to create steam to drive a secondary steam turbine. This is called a combined-cycle topping system. • The second type of system burns fuel (any type) to produce high-pressure steam that then passes through a steam turbine to produce power with the exhaust providing low pressure process steam. This is a steam-turbine topping system • A third type employs heat recovery from an engine exhaust and/or jacket cooling system flowing to a heat recovery boiler, where it is converted to process steam / hot water for further use. • The fourth type is a gas turbine topping system. A natural gas turbine drives a generator. The exhaust gas goes to a heat recovery boiler that makes process steam and process heat.	1+2+ 4	7	7
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XIII	Ten Steps Methodology for Conducting Detailed Energy Audit (as per BIS standard) Ten steps (7 marks)	7	7	7
	PHASE: I 1. a. Plan & organise visit b. Walk through audit of the set up c. Informal interview with production or plant manager or incharge. 2. Conduct a brief meeting / awareness program on energy audit with all divisional heads and persons concerned. PHASE: II 3. Primary data collection, process flow diagram, energy utility diagram and collection of any other relevant datas. 4. Conduct survey and monitoring 5. Conduct detailed trials / experiments for the selected energy guzzlers 6. Analysis of energy use 7. Identification and development of energy conservation opportunities 8. Cost benefit analysis 9. Reporting & presentation to top management on findings PHASE: III 10. Implementation, commissioning and follow up			

XIV	Seven Instruments used for energy audit and mention their purpose. <i>List (Any 7 nos) 7 x 0.5 marks =3.5</i> <i>Purpose (7 x 0.5 marks = 3.5)</i>	7	7	7
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	Temperature Indicator and Measurement device: Temperature indicators are the most important instrument to measure process temperature and heat losses. broadly divide the temperature measurement activity into three measuring instruments: - Infrared non-contact type thermometer, for quick measurement of temperature, this type of thermometer can be used, - Temperature indicator probe type: conventionally temperature indicator with a probe is used for accurately measuring process temperature. along with data logger temperature indicators can be the most useful instrument for an energy audit. - Thermal imager: Thermal imagers are devices and can be used to identify heat loss areas especially for insulation, surface loss, and electrical system. Flue Gas Analyzer: Flue gas analyzer is an important instrument to measure the combustion efficiency of a boiler or furnace. Carbon monoxide, carbon dioxide, and oxygen monitoring is important to identify loss due to incomplete combustion or excess air supply. Power analyzer: Power analyzer is a multi-function power analyzing device which is used to measure precisely direct current, alternating current, AC-voltage, DC-voltage the intensity of DC or AC, phase rotation, and idle, apparent and effective power. Harmonic Analyzer: used to measure frequencies, amplitudes, different phases, and various components of a non-sinusoidal waveform, the harmonic study is important for carrying detailed power quality analysis. Steam Trap Tester/ Monitor: Steam trap traps steam and removes condensate from the steam system. Conductivity,TDS & pH Meter: - Conductivity or TDS meters are required for measuring the salt concentration of the boiler drum or cooling tower. - pH meter helps to identify the acidity or alkalinity of water.			
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Scoring Indicators**Model Question Paper II****ENERGY CONSERVATION AND AUDIT**

	Q. No.	Scoring Indicators	Split score	Sub Total	Tot al score
		PART A			
	I. 1	1.Undertake voluntary initiatives to promote energy conservation 2.Launch and maintain state specific website addressing the voluntary and mandatory provisions of EC Act	1	1	1
	I. 2	Minimum Energy Performance Standards (MEPS) specify the minimum level of energy performance that appliances, lighting and electrical equipment (products) must meet or exceed before they can be offered for sale or used for commercial purposes.	1	1	1
	I. 3	A higher star rating means more a more efficient appliance, which means a lower electricity bill every month. Most modern households have multiple appliances that consume a lot of electricity such as air conditioners, refrigerators, and even ceiling fans.	1	1	1
	I. 4	Main Panel or Main Distribution Panel	1	1	1
	I. 5	● Cogeneration reduce air emissions of particulate matter, nitrous oxides, Sulphur dioxide, mercury and carbon dioxide which would otherwise leads to greenhouse effect. ● It reduces cost of production and improve productivity.	1	1	1
	I. 6	Steam turbines. Gas turbines	1	1	1
	I. 7	Energy audit means the verification, monitoring and analysis of use of energy including submission of technical report containing recommendations for improving energy efficiency with cost benefit analysis and an action plan to reduce energy consumption.	1	1	1
	I. 8	An instrument for measuring the pressure acting on a column of fluid to measure the draft pressure of the combustion system, flue gas path, fans, etc.	1	1	1

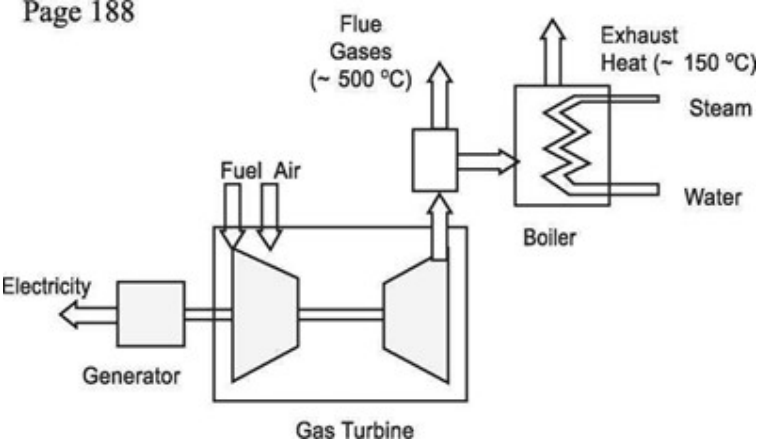
	I. 9	The Sankey diagram is a very useful tool to represent an entire input and output energy flow in any energy equipment or system such as boiler generation, fired heaters, furnaces after carrying out energy balance calculation.	1	1	1
		PART B			

II. 1	<i>Diagram(1)+stages(2)</i> Assessment The foundation of the pyramid is assessment. Assessment is an essential element, not only at the beginning of an energy decision process, but throughout the process. Assessment helps to ensure that any actions taken are an appropriate solution to a problem. Cost and Complexity The pyramid shows that cost and complexity of energy actions typically increase as you move from energy conservation activities, to energy efficiency measures, and finally to alternative energy projects. Conservation and Efficiency Conservation and efficiency measures have numerous benefits by themselves, but are especially important when considering an alternative energy system. The size of an alternative energy system is based on your current energy consumption. Take steps to conserve and use energy resources more efficiently and you will likely be able to install a smaller system that costs less to purchase and operate.	1+2	3	3
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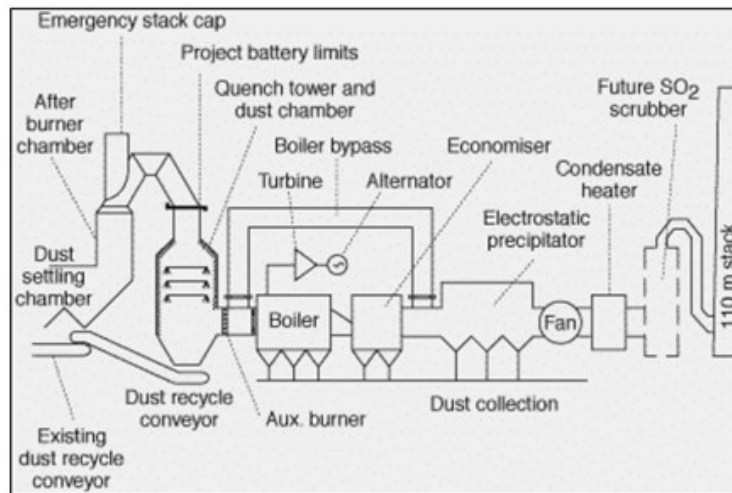
II. 2	<i>Definition (1)+ features(2)</i> Demand-side management or DSM, as applied to energy efficiency measures that modify or reduce end-users' energy demand. This has traditionally been applied to electricity loads but is also used for changes that can be made to demands for all types of energy. Various reasons are put forward for promoting DSM are, • Cost reduction—many DSM and energy efficiency efforts have been introduced in the context of integrated resource planning and aimed at reducing total costs of meeting energy demand; • Environmental and social improvement—energy efficiency and DSM may be pursued to achieve environmental and/or social goals by reducing energy use, leading to reduced greenhouse gas emissions; • Reliability and network issues—ameliorating and/or averting problems in the electricity network through reducing demand in ways which maintain system reliability in the immediate term and over the longer term defer the need for network augmentation; • Improved markets—short-term responses to electricity market conditions (“demand response”), particularly by reducing load during periods of high market prices caused by reduced generation or network capacity.	1+2	3	3

II. 3	<i>Any three (3)</i> ●At starting, the AC Induction motor develops more torque than is required at full speed. ●This stress is transferred to the mechanical transmission system resulting in excessive wear and premature failure of chains, belts, gears, mechanical seals, etc. ●Also, rapid acceleration has a massive impact on electricity supply charges with high inrush currents drawing much higher than that of the normal run current. ●Soft starter provides a reliable and economical solution to these problems by delivering a controlled release of power to the motor, thereby providing smooth, stepless acceleration and deceleration. Motor life will be extended as damage to windings and bearings is reduced. ●Soft Start & Soft Stop is built into 3 phase units, providing controlled starting and stopping with a selection of ramp times and current limit settings to suit all applications	1x3	3	3
II. 4	<i>Any 6 points(6x.5)</i> ● Improve Quality of power supply ● Improve controls ● Improve Power factor ● Improve cooling ● Turn motors off if not in use ● Schedule Regular maintenance ● Provide Regular lubrication ● Reduce Peak Demand by rescheduling motor operation	6x.5	3	3

II. 5	<i>Digram(1.5)+explanation(1.5)</i> The Interline Power Flow Controller (IPFC) is one of the latest generation Flexible AC Transmission Systems (FACTS) controller used to control power flows of multiple transmission lines. Interline power flow controller is a versatile device can be used to control power flows of a multi-line system or sub- networks. The IPFC is a device with the capability of controlling both active and reactive powers between the transmission lines, meaning that voltage phase angle and magnitude are under control	1.5+ 1.5	3	3
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II. 6	<i>Diagram(2)+explanation(1)</i> Gas turbine cogeneration systems can produce all or a part of the energy requirement of the site, and the energy released at high temperature in the exhaust stack can be recovered for various heating and cooling applications. Page 188  Though natural gas is most commonly used, other fuels such as light fuel oil or diesel can also be employed. The typical range of gas turbines varies from a fraction of a MW to around 100 MW.	2+1	3	3
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In a bottoming cycle, the primary fuel produces high temperature thermal energy and the heat rejected from the process is used to generate power through a recovery boiler and a turbine generator. Bottoming cycles are suitable for manufacturing processes that require heat at high temperature in furnaces and kilns, and reject heat at significantly high temperatures. Typical areas of application include cement, steel, ceramic, gas and petrochemical industries. Bottoming cycle plants are much less common than topping cycle plants



The Figure illustrates the bottoming cycle where fuel is burnt in a furnace to produce synthetic rutile. The waste gases coming out of the furnace is utilized in a boiler to generate steam, which drives the turbine to produce electricity.

II. 8	<i>Any three(1x3)</i> ● The Energy Audit provides the vital information base for overall energy conservation program covering essentially energy utilization analysis and evaluation of energy conservation measures. ● It provides recommendations for improving energy efficiency with cost benefit analysis and an action plan to reduce energy consumption. ● Energy audit will help to understand more about the ways energy is used in the industry, and help in identifying the areas where waste can occur and where scope for improvement exists. ● In general, energy audit is the translation of conservation ideas into realities, by evolving technically feasible solutions with economic and other organizational considerations within a specified time.	1x3	3	3
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II.9	<i>Types(1)+explanation(2)</i> Energy Audit can be classified into the following two types i) Preliminary Audit ii) Detailed Audit • Preliminary Energy Audit Methodology Preliminary energy audit is a relatively quick exercise to: • Establish energy consumption in the organization for saving • Identify the most likely (and the easiest areas for attention • Identify immediate (especially no-/low-cost) improvements/ savings • Set a ‘reference point’ • Identify areas for more detailed study/measurement • Preliminary energy audit uses existing, or easily obtained data. Detailed Energy Audit Methodology. A comprehensive audit provides a detailed energy project implementation plan for a facility, since it evaluates all major energy using systems. This type of audit offers the most accurate estimate of energy savings and cost. It considers the interactive effects of all projects, accounts for the energy use of all major equipment, and includes detailed energy cost saving calculations and project cost. In a comprehensive audit, one of the key elements is the energy balance. This is based on an inventory of energy using systems, assumptions of current operating conditions and calculations of energy use. This estimated use is then compared to utility bill charges. Detailed energy auditing is carried out in three phases: Phase I, II and III.	1+2	3	3
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II.10	<i>Diagram (1) + explanation (2)</i> ● Sankey diagrams are flow charts used to visualize material and energy flows with arrows that have a width proportional to the flow quantity. The diagram illustrates the distribution of energy flow from a single input on the left. The input splits into four main paths, each represented by an arrow whose width corresponds to its percentage of the total flow. The paths are: 25% Effective Power (green arrow pointing right), 5% Friction and Parasitic Losses (red arrow pointing up and right), 30% Coolant Loss (red arrow pointing down and right), and 40% Exhaust Gas Loss (red arrow pointing down and right). The labels for the losses are grouped together on the right side of the diagram. <table><tr><th>Flow Component</th><th>Percentage</th></tr><tr><td>Effective Power</td><td>25%</td></tr><tr><td>Friction and Parasitic Losses</td><td>5%</td></tr><tr><td>Coolant Loss</td><td>30%</td></tr><tr><td>Exhaust Gas Loss</td><td>40%</td></tr></table> ● The Sankey diagram is a very useful tool to represent an entire input and output energy flow in any energy equipment or system such as boiler generation, fired heaters, furnaces after carrying out energy balance calculation. ● Usually, the flows are represented by arrows. The width of the arrows is proportional to the size of the actual flow. ● Better than numbers, tables or descriptions, this diagram represents visually various outputs (benefits) and losses so that energy managers can focus on finding improvements in a prioritized manner ● From the above diagram it can be found that Exhaust flue gas losses are a key area for priority attention (40 %)	Flow Component	Percentage	Effective Power	25%	Friction and Parasitic Losses	5%	Coolant Loss	30%	Exhaust Gas Loss	40%	1+2	3	3
Flow Component	Percentage													
Effective Power	25%													
Friction and Parasitic Losses	5%													
Coolant Loss	30%													
Exhaust Gas Loss	40%													

	PART C			
III	<i>Any seven(1x7)</i> The salient features of energy conservation act 2001 are EC act empowers the union government and in some instances the state government to ● Notify energy-intensive industries, establishments and commercial buildings as designated consumers ● Prescribe energy consumption norms and standards for designated consumers ● Direct designated consumers to appoint certified energy managers for efficient use of energy ● State government to amend the energy conservation building codes to suit regional and climatic conditions ● Direct owners of commercial building to comply with the energy conservation building codes ● Direct mandatory display of labels on notified equipment and appliances ● Specify energy consumption standards for notified equipment and appliances ● Prohibit manufacture, sale, purchase and import of notified equipment and appliances not conforming to standards ● Under the provision of this Act the Bureau of Energy Efficiency (BEE) was established with effect from 1.03.2002 ● The mission of the BEE is to institutionalize energy efficiency services, promote energy efficiency delivery mechanisms and provide leadership for improvement of energy efficiency in all sectors of the economy.	1x7	7	7
	OR			

IV	<i>Definition (1) + Any six needs (6)</i> Energy conservation is the effort made to reduce the consumption of energy by using less of an energy service. This can be achieved either by using energy more efficiently (using less energy for a constant service) or by reducing the amount of service used (for example, by driving less). Most of our energy use comes from fossil fuels like petroleum and coal that provide electricity and gas to power our growing energy needs. These resources are non-renewable which means that we will eventually run out. Conserving energy not only helps to conserve resources but also translates into financial savings. Needs: ● We use energy faster than it can be produced. Coal, oil and natural gas - the most utilized sources take thousands of years for formation. ● Energy resources are limited. India has approximately 1% of world's energy resources but it has 16% of the world population. ● Most of the energy resources we use cannot be reused and renewed. Non - renewable energy resources constitute 80% of the fuel use. It is said that our energy resources may last only for another 40 years or so. ● We save the country a lot of money when we save energy. About 75% of our crude oil needs are met from imports which would cost about 1, 50, 000 crores per year. ● Energy saved is energy generated. When we save one unit of energy, it is equivalent to 2 units of energy produced. ● Save energy to reduce pollution. Energy production and use account to large proportion of air pollution and more than 83 percent of greenhouse gas emissions.	1+6	7	7
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V	<i>Any seven points(1x7)</i> Role of State Designated Agencies As per Energy Conservation Act 2001, State Governments have been empowered to designate agencies (State Designated Agency, i.e. SDA) in consultation with the Bureau of Energy Efficiency. Designated Agencies will have the responsibility to implement the Act within the State. <i>Responsibilities</i> ● Spread awareness on EC Act ● Undertake voluntary initiatives to promote energy conservation ● Liaison and coordinate with BEE, State Government Departments dealing with energy, industry, planning, regulators, consumer affairs, municipal bodies etc. ● Capacity building of staff employed ● Launch and maintain state specific website addressing the voluntary and mandatory provisions of EC Act ● Undertake energy conservation awareness program for consumers, industrial & commercial sector, school children, farmers etc. ● Arrange interactive meets between energy managers, energy auditors, designated consumers and other experts	1x7	7	7
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V1	<i>Explanation (2) +benefits (5)</i> The total installed generating capacity in the country is over 1,56,700 Approx. 28000 MW (i.e. 17 %) is lighting load. There are no mandatory policies or regulations for the adoption of CFLs in households in India. Some of the progressive utilities in India have tried out bulk procurement and marketing of CFLs to households at below market prices and / or introducing replacement schemes. –could not sustain. National Mission for Enhanced Efficiency (NMEEE) under the National Action Plan for Climate Change (NAPCC) also prompts the BEE to accelerate the shift towards energy efficient appliances. The Bachat Lamp Yojana (BLY) scheme is an indirect outcome of the above policy scenario. Hence, the BLY objective is to – replace ICLs amongst residential users in India with quality long-life CFLs. Benefits ● Reduced Peak Load Demand, less load shedding and power-cuts. ● Reduced installation ratings, thus lowering peak demand reduces overall plant and Capital cost requirements. ● Reduced Electricity Bills ● Less Load shedding ● High Returns on the investment from the CERs generated.	2+5	7	7
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VII	<i>Any seven points(1x7)</i> ● Installation of energy efficient fluorescent lamps in place of “Conventional” fluorescent lamps. ● Installation of Compact Fluorescent Lamps (CFL's) in place of incandescent lamps. ● Installation of metal halide lamps in place of mercury / sodium vapour lamps ● Installation of High-Pressure Sodium Vapour (HPSV) lamps for applications where colour rendering is not critical. ● Installation of LED panel indicator lamps in place of filament lamps. ● Grouping of lighting system, to provide greater flexibility in lighting control ● Installation of microprocessor-based controllers. ● Optimum usage of daylighting ● Installation of "exclusive" transformer for lighting	1x7	7	7
	OR			

VIII	<i>Diagram (2) + two equipment explanation (5)</i> 1.A variable-frequency drive (VFD) is a system for controlling the rotational speed of an alternating current (AC) electric motor by controlling the frequency of the electrical power supplied to the motor. Variable-frequency drives are also known as adjustable-frequency drives (AFD), variable speed drives (VSD), AC drives, micro-drives or inverter drives. Since the voltage is varied along with frequency, these are sometimes also called VVVF (variable voltage variable frequency) drives. A variable frequency drive system generally consists of an AC motor (mostly 3 phase induction motor), a controller (solid state electronic power conversion devices like diodes, thyristors, IGBTs,etc.) and an operator interface. 2. Automatic star delta converter. • All motors of higher capacity are provided with star/delta windings. • During the initial startup, the motor is connected in star mode to reduce the initial inrush of current. • Once the motor achieves the required speed, it switches to the delta mode, after which the motor can be loaded. • ADLS makes use of the fact that whenever the load on the motor is less (<40%), it switches the motor to star mode, and as soon as the load increases (>40%), it switches the motor to delta mode. • When the motor operates in the star mode its current consumption drops by 1/3, thereby saving energy. • The saving is near about 10-40% depending on the motor load and the load changes.	2+5	7	7
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IX	<i>Concept (2) +any 5 techniques (5)</i> ● Energy-efficient motors (EEM) are the ones in which, design improvements are incorporated specifically to increase operating efficiency over motors of standard design. ● Design improvements focus on reducing intrinsic motor losses. Improvements include the use of lower-loss silicon steel, a longer core (to increase active material), thicker wires (to reduce resistance), thinner laminations, smaller air gap between stator and rotor, copper instead of aluminum bars in the rotor, superior bearings and a smaller fan, etc. ● Energy-efficient motors now available in India operate with efficiencies that are typically 3 to 4 percentage points higher than standard motors ● In keeping with the stipulations of the BIS, energy-efficient motors are designed to operate without loss in efficiency at loads between 75 % and 100 % of rated capacity. This may result in major benefits in varying load applications. The power factor is about the same or may be higher than for standard motors. Techniques for improving efficiency 1. Iron -Use of thinner gauge, lower loss core steel reduces eddy current losses. Longer core adds more steel to the design, which reduces losses due to lower operating flux densities. 2. Stator I²R Use of more copper and larger conductors increases cross sectional area of stator windings. This lowers resistance (R) of the windings and reduces losses due to current flow (I). 3. Rotor I²R Use of larger rotor conductor bars increases size of cross section, lowering conductor resistance (R) and losses due to current flow (I). 4. Friction & Windage Use of low loss fan design reduces losses due to air movement 5. Stray Load Loss Use of optimized design and strict quality control procedures minimizes stray load losses.			
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X	<i>Explanation any seven points(1x7)</i> ● Energy efficient transformers use an amorphous core-with unique physical and magnetic properties. ● These transformers have increased efficiency even at low loads - 98.5% efficiency at 35% load. ● It provides an excellent opportunity to conserve energy right from the installation. ● These transformers are a little costlier than conventional iron core transformers, ● The expected reduction in core loss over conventional (Si Fe core) transformers is roughly around 70%, which is quite significant. ● The amorphous metal transformer is an example. ● Energy efficient transformers are therefore an important means to reduce transmission and distribution loss.	1x7	7	7
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XI

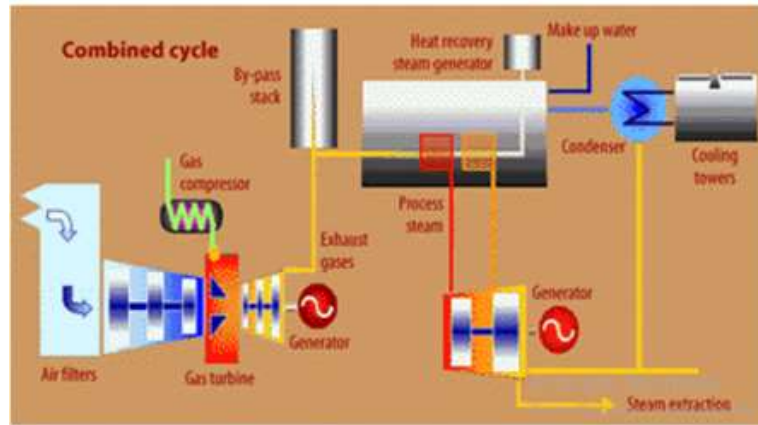
Diagram (2+2) + explanation (1.5+1.5)

4+3

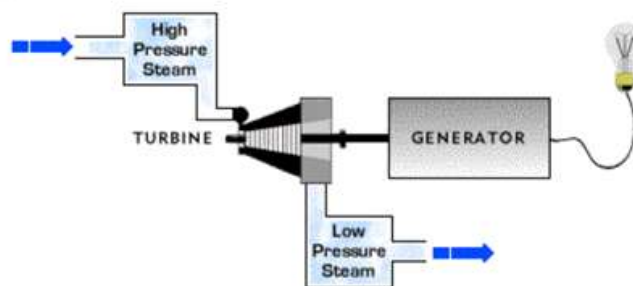
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A gas turbine or diesel engine producing electrical or mechanical power followed by a heat recovery boiler to create steam to drive a secondary steam turbine. This is called a combined-cycle topping system.



The second type of system burns fuel (any type) to produce high-pressure steam that then passes through a steam turbine to produce power with the exhaust provides low pressure process steam. This is a steam-turbine topping system.



XII	<i>Any seven points with explanation(1x7)</i> Electrical load matching: • In the base electrical load matching, size of cogeneration plant is chosen to fulfill minimum demand curve. The remaining power need is satisfied by purchasing from the utility. Thermal load matching: • In base thermal load matching, minimum heat requirement of the premises is considered while sizing of the cogeneration plant. auxiliary boilers can supply heat in case of additional heating needs. If excess thermal energy is available, it can be exported to the needy places. Thermal to electrical energy ratio: • This is also called as heat to power ratio. • Heat to power ratio is one of the most influential technical parameters which can govern the selection of cogeneration system. Quality of required thermal energy: • Quality of thermal energy refers to the temperature and pressures of the steam which can be an attributing factor for the selection of cogeneration scheme. Fuel availability: • The type of readily available fuel is also an important consideration while choosing a cogeneration system. Almost every fuel can be used by a system having compatibility with the same. Installation and available space: • Cogeneration system may depend upon whether it's a retrofit or new installation. Grid-connected/independent system: • Technical design of cogeneration plant may depend upon whether it's connected to grid or it's a standalone system. Operational costs:			
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XIII	<i>Explanation (4) +opportunities (3)</i> The energy audit for building The study should reveal to the owner or management team of the building the options available for reducing energy waste, the costs involved and benefits achievable from implementing those energy conservation opportunities The energy audit may range from a simple walk-through survey at one extreme to one that may span several phases. These phases include a simple walk-through survey followed by monitoring of energy use in the building services and then model analysis using computer simulation of building operation The first stage is to reduce energy use in areas where energy is wasted and reductions will not cause disruptions to various functions. The level of service must not be compromised by the reduction in energy consumed. It begins with a detailed step by step analysis of the building's energy use factors and costs, such as insulation values, occupancy schedules, chiller efficiencies, lighting levels and records of utility and fuel expenditures The second stage is to improve efficiency of energy conversion equipment and to reduce energy use by proper operations and maintenance. Here the energy conservation opportunities include the following ● Building equipment operation ● Building envelope ● Air conditioning and mechanical ventilation equipment and systems ● Lighting systems ● Power systems and ● Miscellaneous services The last stage is to carry out large scale energy reducing measures when existing facilities have past their useful life or require extensive repairs or replacement because of obsolescence. In this case, higher energy savings may be achieved.	4+3	7	7
	OR			

XIV	After successfully carried out energy audit, the energy manager/energy auditor should report to the top management for effective communication and implementation. A typical energy audit reporting contents and format are given below. The following format is applicable for most of the industries. However, the format can be suitably modified for specific requirement applicable for a particular type of water. industry. ● Acknowledgement ● Executive summary - Energy audit options at a glance and recommendations ● Introduction about the plant ● Production process description ● Energy and utility system description - List of utilities - Brief description of each utility ● Detailed process flow diagram and energy and material balance ● Energy efficiency in utility and process systems ● Energy conservation options and recommendations - List of options in terms of no cost / low cost, medium cost, and high investment cost, annual energy and cost savings, and pay back - Implementation plan for energy saving measures / projects ● Annexures	7	7	7
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